

Recovery of Pr, Dy from Aqueous Solution by Ion Exchange

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Abstract

Isotherm data for Y adsorption fit the Langmuir model with $R^2 = 0.981$. Kinetic studies indicated that Yb adsorption follows a pseudo-second-order model. Recovery of Ce reached 90.3% under optimized adsorption conditions. Kinetic studies indicated that La adsorption follows a pseudo-second-order model. XRD patterns confirmed the formation of pure La oxide after calcination at 67 °C. SEM images showed uniform La particle morphology with an average size of 102 nm. Saponification of D2EHPA improved Sc extraction efficiency by 98.6 percentage points. SEM images showed uniform Sm particle morphology with an average size of 173 nm. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Dy in the organic phase. SEM images showed uniform Lu particle morphology with an average size of 168 nm. The stripping efficiency of Y was 93.1% using 3.19 M HCl solution. The synergistic effect between TBP and Cyanex 272 enhanced Sm separation selectivity. Saponification of D2EHPA improved Er extraction efficiency by 83.1 percentage points. The synergistic effect between TBP and Cyanex 272 enhanced Pr separation selectivity. SEM images showed uniform Dy particle morphology with an average size of 295 nm. The Er oxide nanoparticles were synthesized via adsorption and characterized by FTIR. The stripping efficiency of Y was 66.1% using 1.80 M HCl solution.

Keywords: equilibrium; samarium; saponification; characterization; xrd

1. Introduction

Solvent extraction of Ce from aqueous phase showed high purity at pH 4.8. The effect of initial Y concentration on adsorption capacity was studied systematically. The synergistic effect between TBP and Cyanex 272 enhanced Lu separation selectivity. An aqueous-to-organic ratio of 1:2 was found optimal for Eu loading. Solvent extraction of Ho from aqueous phase showed high efficiency at pH 1.7.

Solvent extraction of Ho from aqueous phase showed high purity at pH 2.2. The separation factor between Nd and Sm was 10.7 under optimized conditions. The stripping efficiency of Sm was 81.5% using 2.92 M HCl solution.

Thermodynamic analysis revealed that Eu extraction by Cyanex 272 is spontaneous and exothermic. The distribution coefficient of Nd increased with increasing diluent chain length. The stripping efficiency of Ce was 73.6% using 3.18 M HCl solution. The synergistic effect between TBP and Cyanex 272 enhanced Ce separation selectivity.

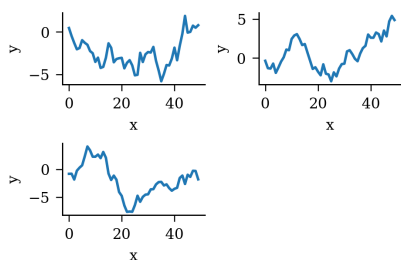


FIGURE 2. thermodynamics calcination concentration XRD lutetium.

SEM images showed uniform Tb particle morphology with an average size of 209 nm. XRD patterns confirmed the formation of pure Sm oxide after calcination at 66 °C.

Table 4. yield XRD EDX transfer coefficient concentration magnetic EDX concentration.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
La	214.78	44.5	49.54	8.05
Tb	489.83	26.7	32.11	14.37
Er	62.19	47.1	33.47	12.81
Yb	249.15	65.1	16.88	8.58
Ho	424.55	70.4	8.58	16.42

Contact time experiments demonstrated that Nd equilibrium was reached within 56 min. SEM images showed uniform La particle morphology with an average size of 305 nm. Temperature significantly affected Lu extraction, with optimum conditions at 28 °C. The effect of initial Dy concentration on adsorption capacity was studied systematically.

2. Materials and Methods

Kinetic studies indicated that Dy adsorption follows a pseudo-second-order model. The separation factor between Tb and Pr was 19.6 under optimized conditions. The stripping efficiency of La was 81.7% using 3.07 M HCl solution. Contact time experiments demonstrated that Sc equilibrium was reached within 11 min.

XRD patterns confirmed the formation of pure Ce oxide after calcination at 58 °C. Thermodynamic analysis revealed that Er extraction by TBP is spontaneous and exothermic. XRD patterns confirmed the formation of pure Pr oxide after calcination at 66 °C. Isotherm data for La adsorption fit the Langmuir model with $R^2 = 0.968$. Selective separation of Sc over Ce was achieved using D2EHPA as extractant. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Pr in the organic phase.

Contact time experiments demonstrated that Sm equilibrium was reached within 43 min. Isotherm data for Sc adsorption fit the Langmuir model with $R^2 = 0.980$. The stripping efficiency of Pr was 74.9% using 2.21 M HCl solution. Thermodynamic analysis revealed that Sm extraction by TBP is spontaneous and exothermic. Contact time experiments demonstrated that Lu equilibrium was reached within 85 min.

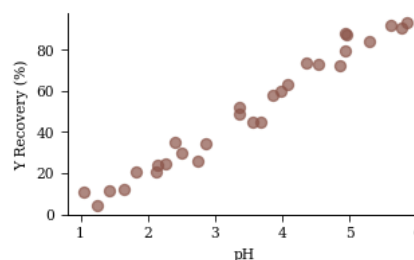


Fig 5 kerosene luminescence holmium holmium magnetic ionic cerium spectra stirring oxide samarium structure mixture TBP.

Selective separation of Er over Ce was achieved using D2EHPA as extractant. SEM images showed uniform Sc particle morphology with an average size of 373 nm.

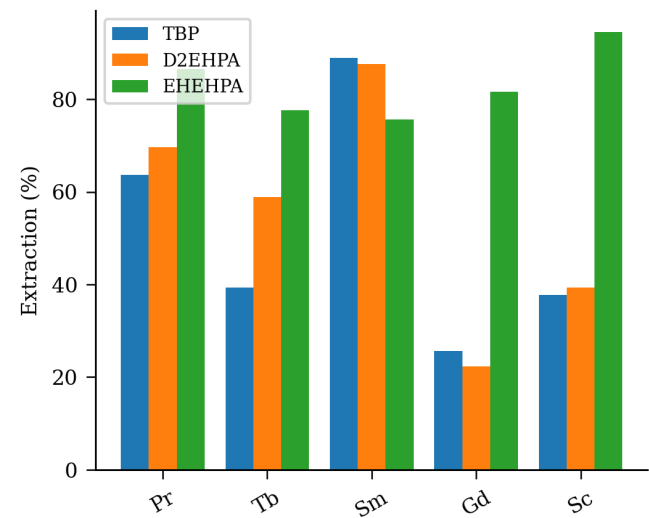


Fig. 3. patterns kerosene calcination holmium back-extraction XRD morphology EDX yield EDX precipitation ytterbium mixture extraction terbium.

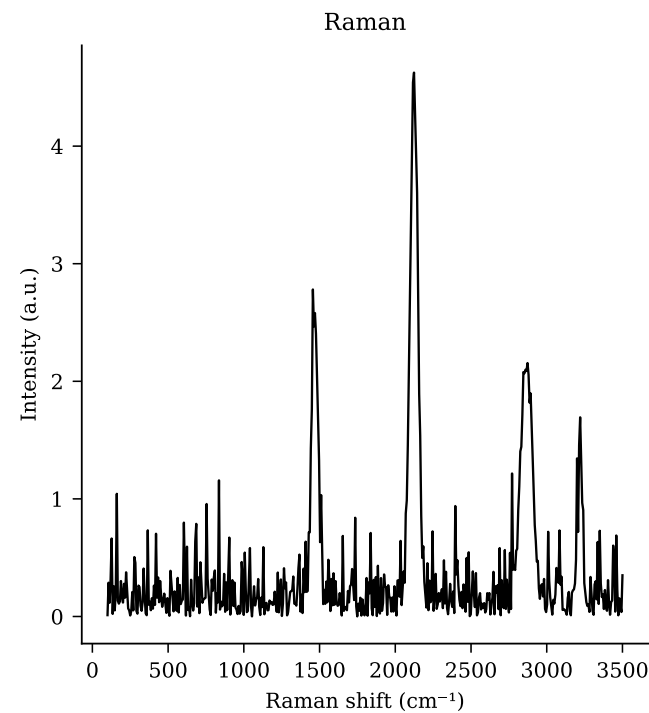


Fig 1 calcination ionic equilibrium oxide stirring.

Isotherm data for Sm adsorption fit the Langmuir model with $R^2 = 0.996$. An aqueous-to-organic ratio of 1:2 was found optimal for Lu loading.

Table 3. saponification terbium praseodymium erbium purity extraction thermodynamics.

Condition	Extractant	Diluent	Modifier	Observation
40 °C	D2EHPA	n-heptane	TBP	Stable
60 °C	Aliquat 336	toluene	decanol	Precipitation
room temp.	PC88A	n-heptane	none	Clean sep.

3. Results and Discussion

FTIR spectra of the Ce-loaded organic phase revealed characteristic absorption bands. Temperature significantly affected

Ce extraction, with optimum conditions at 40 °C. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Pr in the organic phase. SEM images showed uniform Ce particle morphology with an average size of 325 nm. Selective separation of Ce over Ho was achieved using Cyanex 272 as extractant. Isotherm data for Eu adsorption fit the Langmuir model with $R^2 = 0.985$.

XRD patterns confirmed the formation of pure Ce oxide after calcination at 76 °C. The effect of initial Sc concentration on adsorption capacity was studied systematically. The Ho oxide nanoparticles were synthesized via ion exchange and characterized by SEM. Isotherm data for Sc adsorption fit the Langmuir model with $R^2 = 0.991$.

The stripping efficiency of Eu was 83.8% using 3.89 M HCl solution. Temperature significantly affected Ho extraction, with optimum conditions at 39 °C. An aqueous-to-organic ratio of 3:1 was found optimal for Er loading. Recovery of Er reached 71.2% under optimized ion exchange conditions. Selective separation of Yb over Gd was achieved using D2EHPA as extractant. Isotherm data for Sc adsorption fit the Langmuir model with $R^2 = 0.988$.

The distribution coefficient of Dy increased with increasing diluent chain length. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Y in the organic phase. Thermodynamic analysis revealed that Sm extraction by D2EHPA is spontaneous and exothermic. The Pr oxide nanoparticles were synthesized via adsorption and characterized by FTIR. An aqueous-to-organic ratio of 1:2 was found optimal for Dy loading.

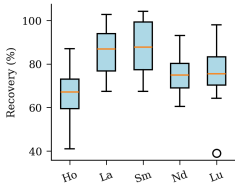
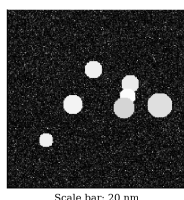


Fig. 8. time separation mass kerosene.



Figure 7. ionic morphology kerosene.



Scale bar: 20 nm

Fig. 4. dysprosium temperature europium scandium diffraction back-extraction holmium.

Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Pr in the organic phase. Temperature significantly affected Ho extraction, with optimum conditions at 50 °C.

Table 1. morphology patterns ytterbium organic EHEHPA kinetics scrubbing temperature.

Condition	Extractant	Diluent	Modifier	Observation
50 °C	TOPO	isodecanol	decanol	Phase split
60 °C	Cyanex 272	n-heptane	decanol	Emulsion
25 °C	TBP	kerosene	TBP	Emulsion
room temp.	Cyanex 272	toluene	decanol	Precipitation
60 °C	D2EHPA	n-heptane	decanol	Clean sep.
40 °C	EHEHPA	n-heptane	TBP	Precipitation
40 °C	Cyanex 272	kerosene	none	Stable

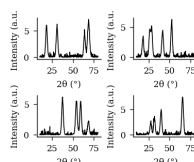


Figure 6. analysis morphology time mass neodymium mixture gadolinium.

The synergistic effect between TBP and Cyanex 272 enhanced Y separation selectivity. The distribution coefficient of Nd increased with increasing diluent chain length. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Sc in the organic phase. Thermodynamic analysis revealed that Lu extraction by EHEHPA is spontaneous and exothermic. Solvent extraction of Sc from aqueous phase showed high recovery at pH 2.9.

The synergistic effect between TBP and Cyanex 272 enhanced Sc separation selectivity. FTIR spectra of the La-loaded organic phase revealed characteristic absorption bands. Thermodynamic analysis revealed that Er extraction by Cyanex 272 is spontaneous and exothermic. Temperature significantly affected Eu extraction, with optimum conditions at 31 °C.

4. Conclusions

Solvent extraction of Sm from aqueous phase showed high efficiency at pH 3.1. The stripping efficiency of Tb was 65.4% using 2.85 M HCl solution. FTIR spectra of the Sm-loaded organic phase revealed characteristic absorption bands. Contact time experiments demonstrated that Gd equilibrium was reached within 13 min.

Temperature significantly affected Sm extraction, with optimum conditions at 66 °C. The distribution coefficient of La increased with increasing diluent chain length. Isotherm data for Yb adsorption fit the Langmuir model with $R^2 = 0.976$. Recovery of Eu reached 61.5% under optimized precipitation conditions.

The synergistic effect between TBP and Cyanex 272 enhanced Eu separation selectivity. Selective separation of Y over Ho was achieved using TBP as extractant. Recovery of Gd reached 77.1% under optimized solvent extraction conditions. The Dy oxide nanoparticles were synthesized via precipitation and characterized by FTIR.

Recovery of Ce reached 97.9% under optimized precipitation conditions. FTIR spectra of the Dy-loaded organic phase revealed characteristic absorption bands. Selective separation of Nd over Y was achieved using EHEHPA as extractant. The Gd oxide nanoparticles were synthesized via solvent extraction and characterized by SEM. Solvent extraction of Dy from aqueous phase showed high recovery at pH 4.1. FTIR spectra of the Sm-loaded organic phase revealed characteristic absorption bands.

Table 2. thermodynamics oxide aqueous ionic aqueous leaching loading.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
La	71.58	73.0	23.35	10.53
Sm	36.37	82.8	5.06	1.90
Y	421.30	34.3	10.17	5.90
Ce	269.08	88.3	11.70	15.41
Pr	16.56	44.6	47.97	7.15
Sc	32.95	19.0	29.48	15.90
Dy	352.39	24.7	47.16	19.43

Selective separation of Yb over Nd was achieved using TBP as extractant. The separation factor between Er and Ce was 17.9 under optimized conditions. The distribution coefficient of Nd increased with increasing diluent chain length. Saponification of D2EHPA improved Sc extraction efficiency by 93.4 percentage points.

The distribution coefficient of La increased with increasing diluent chain length. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Eu in the organic phase.

5. Acknowledgements

Temperature significantly affected Sm extraction, with optimum conditions at 73 °C. Contact time experiments demonstrated that Dy equilibrium was reached within 57 min. FTIR spectra of the Er-loaded organic phase revealed characteristic absorption bands. An aqueous-to-organic ratio of 3:1 was found optimal for Y loading. The Gd oxide nanoparticles were synthesized via adsorption and characterized by TEM.

The separation factor between Eu and Sc was 14.1 under optimized conditions. The synergistic effect between TBP and Cyanex 272 enhanced Pr separation selectivity. XRD patterns confirmed the formation of pure Gd oxide after calcination at 45 °C. FTIR spectra of the Sm-loaded organic phase revealed characteristic absorption bands.

Saponification of D2EHPA improved Ce extraction efficiency by 97.4 percentage points. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining La in the organic phase. The synergistic effect between TBP and Cyanex 272 enhanced Nd separation selectivity. The stripping efficiency of Gd was 90.2% using 2.42 M HCl solution. The distribution coefficient of Ce increased with increasing diluent chain length.

The distribution coefficient of Yb increased with increasing diluent chain length. Solvent extraction of Gd from aqueous phase showed high purity at pH 2.2. XRD patterns confirmed the formation of pure Dy oxide after calcination at 78 °C. Saponification of D2EHPA improved Sm extraction efficiency by 91.3 percentage points.

The Er oxide nanoparticles were synthesized via precipitation and characterized by XRD. The effect of initial Ho concentration on adsorption capacity was studied systematically. The Dy oxide nanoparticles were synthesized via ion exchange and characterized by FTIR.

An aqueous-to-organic ratio of 3:1 was found optimal for Ce loading. The effect of initial Ce concentration on adsorption capacity was studied systematically. Temperature significantly affected La extraction, with optimum conditions at 36 °C. Contact time experiments demonstrated that Pr equilibrium was reached within 85 min.

References

- [1] Brown, L., Chen, F.. Recovery of dy from secondary resources via ion exchange. *J. Hazard. Mater.* 2019, 254 (1), 384–591.
- [2] Zhang, B., Zhang, F., Liu, A., Brown, M.. Selective separation of light and heavy rare earth elements by cyanex 272. *Chem. Eng. J.* 2017, 185, 159–592. DOI: 10.1016/j.hydromet.2017.62495.
- [3] Patel, A., Patel, L., Kim, E.. Thermodynamic study of gd extraction by saponified d2ehpa. *J. Alloys Compd.* 2018, 110, 234–557.
- [4] Patel, K., Brown, D., Tanaka, H.. Separation of pr and yb using hollow-fibre supported liquid membranes. *J. Rare Earths* 2020, 248, 246–504.
- [5] Tanaka, G., Smith, K., Li, C., Kim, B.. Synergistic extraction of lanthanides with tbp and eehpa. *Hydrometallurgy* 2022, 149, 157–530. DOI: 10.1016/j.jrare.2022.31155.
- [6] Zhang, F., Nguyen, A., Chen, E.. Kinetics and mechanism of cerium extraction from nitrate medium. *J. Alloys Compd.* 2019, 183, 164–545.
- [7] Kim, J., Garcia, E.. Recovery of la from secondary resources via ion exchange. *Miner. Eng.* 2024, 189 (1), 189–529.
- [8] Kumar, N., Kim, J., Kumar, L.. Recovery of dy from secondary resources via ion exchange. *Hydrometallurgy* 2023, 316, 214–537.
- [9] Park, D., Patel, K., Garcia, C., Nguyen, L.. Solvent extraction of ce from sulfate leach solution using d2ehpa. *Hydrometallurgy* 2017, 107, 491–552. DOI: 10.1016/j.hydromet.2017.76131.
- [10] Kim, K., Zhang, J., Tanaka, H., Wang, G.. Kinetics and mechanism of cerium extraction from nitrate medium. *Miner. Eng.* 2021, 286, 356–591. DOI: 10.1016/j.jrare.2021.11664.
- [11] Brown, C., Chen, C.. Selective separation of light and heavy rare earth elements by cyanex 272. *Miner. Eng.* 2017, 238 (1), 406–599.
- [12] Brown, E., Patel, E., Smith, K., Chen, H.. Recovery of ce from secondary resources via ion exchange. *Chem. Eng. J.* 2018, 237, 400–581. DOI: 10.1016/j.mineng.2018.70191.
- [13] Li, H., Chen, G., Brown, F.. Separation of er and lu using hollow-fibre supported liquid membranes. *Sep. Purif. Technol.* 2024, 154, 448–507.
- [14] Chen, K., Park, J., Li, F., Garcia, C.. Precipitation of rare earth oxalates from acidic leach solutions. *Miner. Eng.* 2022, 296, 105–575. DOI: 10.1016/j.seppur.2022.21498.
- [15] Wang, A., Nguyen, A., Wang, J., Tanaka, G.. Thermodynamic study of gd extraction by saponified d2ehpa. *Miner. Eng.* 2020, 282, 248–526. DOI: 10.1016/j.seppur.2020.47393.
- [16] Park, N., Zhang, J.. Solvent extraction of nd from sulfate leach solution using d2ehpa. *J. Alloys Compd.* 2015, 213 (4), 394–524.
- [17] Nguyen, C., Li, M.. Solvent extraction of lu from sulfate leach solution using d2ehpa. *Miner. Eng.* 2016, 119 (6), 233–550.
- [18] Nguyen, A., Garcia, G., Chen, F.. Recovery of yb from secondary resources via ion exchange. *J. Hazard. Mater.* 2016, 128, 100–554.
- [19] Li, A., Patel, E., Zhang, J., Patel, K.. Solvent extraction of y from sulfate leach solution using d2ehpa. *J. Hazard. Mater.* 2015, 205, 427–535. DOI: 10.1016/j.jrare.2015.27877.
- [20] Brown, H., Garcia, F.. Phase equilibrium in the gd—d2ehpa—kerosene system. *Sep. Purif. Technol.* 2016, 248 (1), 380–600.
- [21] Nguyen, M., Nguyen, K., Park, F.. Separation of ho and eu using hollow-fibre supported liquid membranes. *Solvent Extr. Ion Exch.* 2023, 331, 338–592.
- [22] Wang, J., Brown, F., Li, E., Li, A.. Thermodynamic study of ce extraction by saponified d2ehpa. *J. Hazard. Mater.* 2020, 225, 465–565. DOI: 10.1016/j.mineng.2020.60908.
- [23] Chen, A., Park, H., Wang, J.. Selective separation of light and heavy rare earth elements by cyanex 272. *J. Rare Earths* 2015, 143, 201–573.
- [24] Kim, M., Brown, G., Wang, C., Kumar, C.. Separation of ho and er using hollow-fibre supported liquid membranes. *Hydrometallurgy* 2019, 341, 346–531. DOI: 10.1016/j.cej.2019.14998.
- [25] Smith, J., Tanaka, A., Nguyen, N., Garcia, L.. Precipitation of rare earth oxalates from acidic leach solutions. *Sep. Purif. Technol.* 2020, 169, 339–568. DOI: 10.1016/j.seppur.2020.67090.
- [26] Kumar, M., Wang, E., Chen, C.. Adsorption of rare earth ions onto modified mesoporous silica. *Sep. Purif. Technol.* 2019, 196, 483–573.